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6

UML Part 2

Sequence Diagrams

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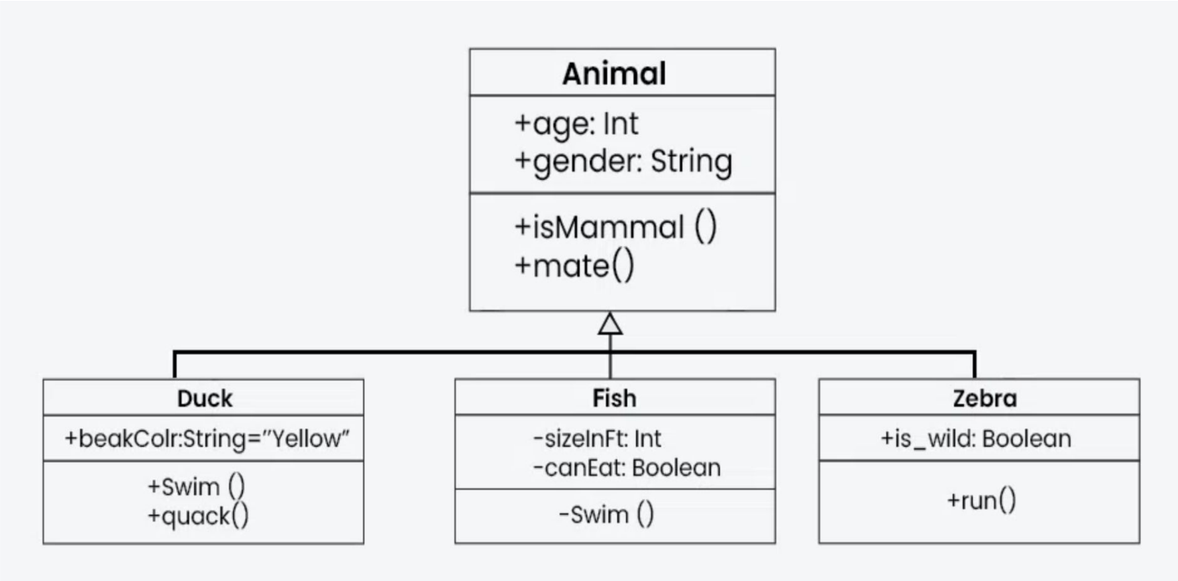
Recap of Previous Lecture!

Quick Recap

1. **Class diagram is structural or behavioural?**

2. **How do we display a class in UML?**

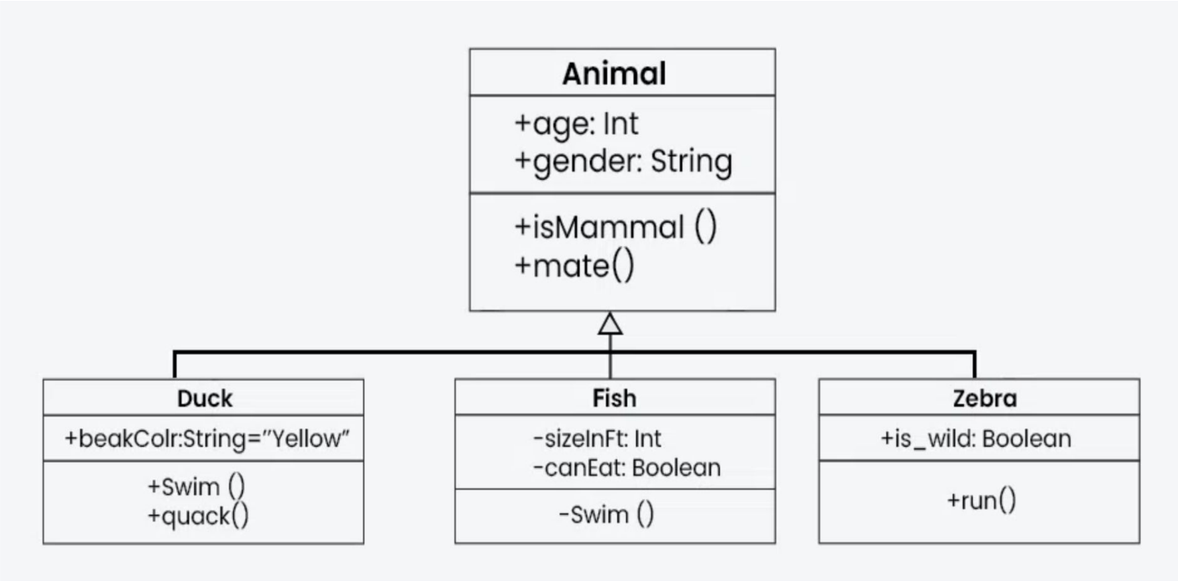
3. **How do you show?**
   - *Attributes*
   - *Behaviour*
   - *Relation*



Quick Recap

Class diagram tells me ___?___

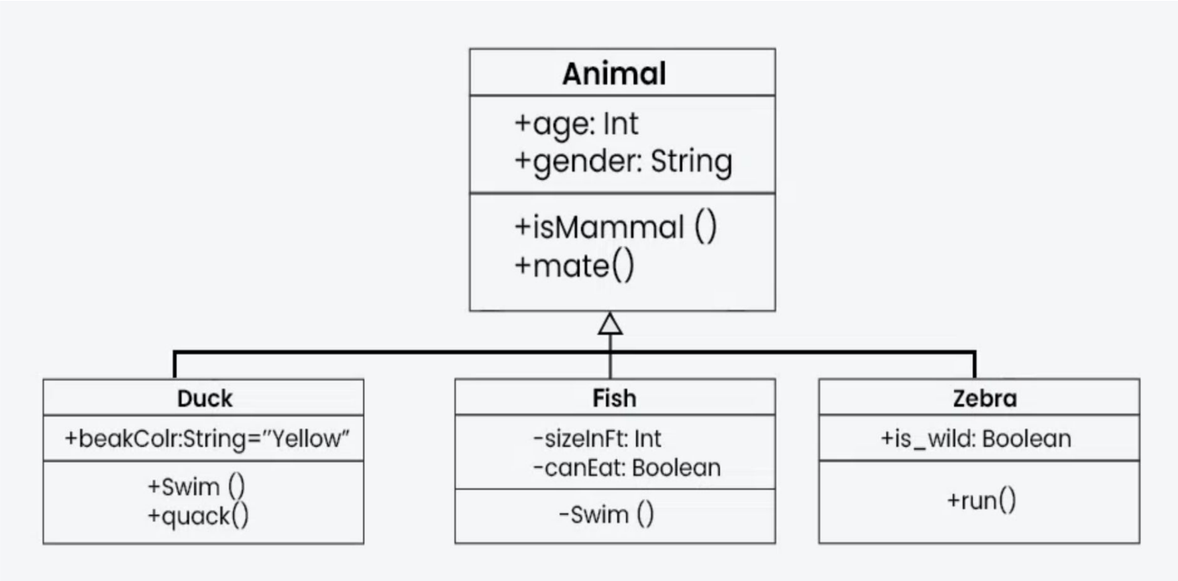
But class diagram does NOT tell me ___?___



Quick Recap

Class diagram tells me **structure**, **relationships**, **attributes**, **methods**.

But class diagram does **NOT** tell me **runtime flow / order of interactions / time sequence**



Class Diagram

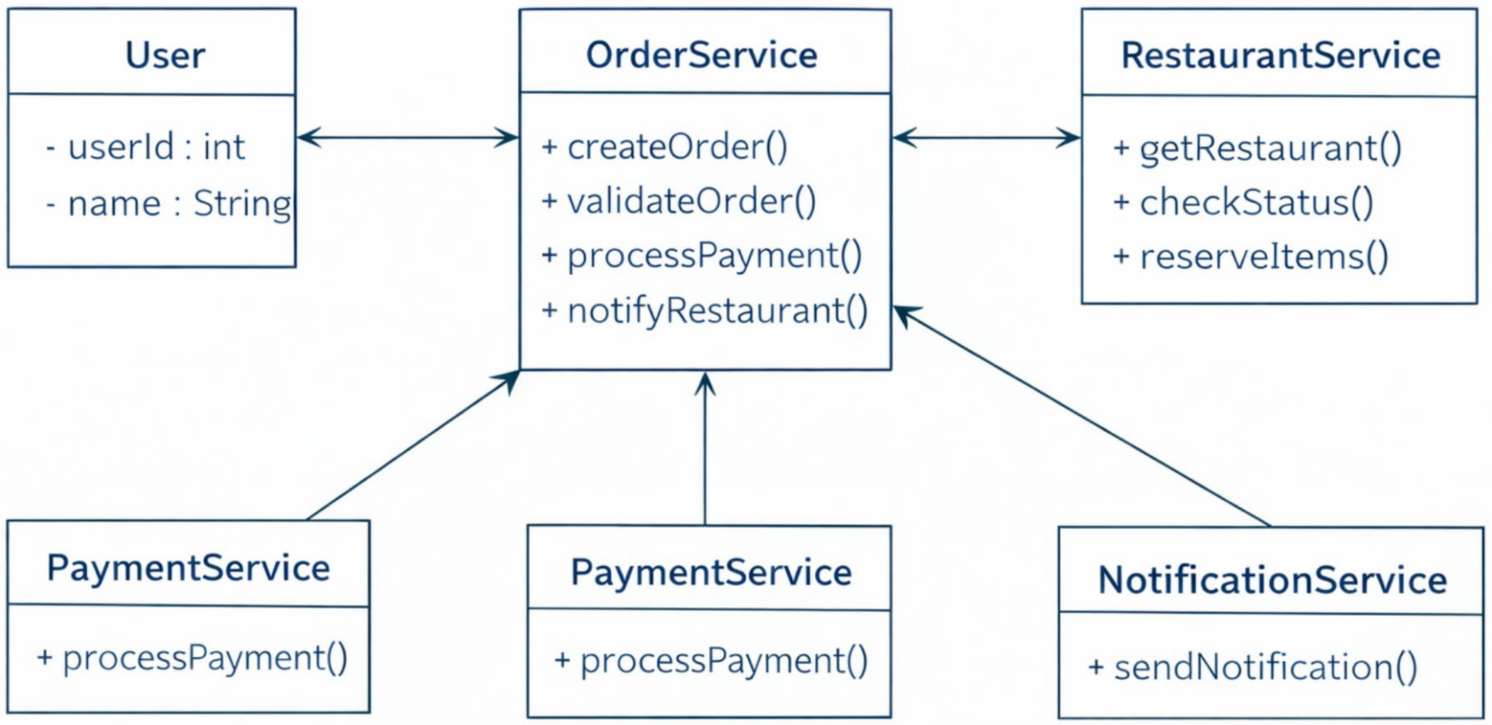
Class diagram = static blueprint

Sequence Diagram

Sequence diagram = runtime story

Sequence Diagram

Here’s a class diagram. Now tell me: when a user starts placing an order, what happens first and then?



Sequence Diagram

Sequence Diagram

All these things happen in a sequence. Right?

The class diagram can’t settle this argument.

Sequence diagram is for the order of interactions.

Sequence Diagram - Components

Actor: User

Lifeline: User / UI / services / DB

Message: placeOrder(), isOpen(), save()

Activation: “busy doing work”

Return: dashed arrow

Fragments:

- **alt** = if/else
- **opt** = optional

Sequence Diagram - Actor

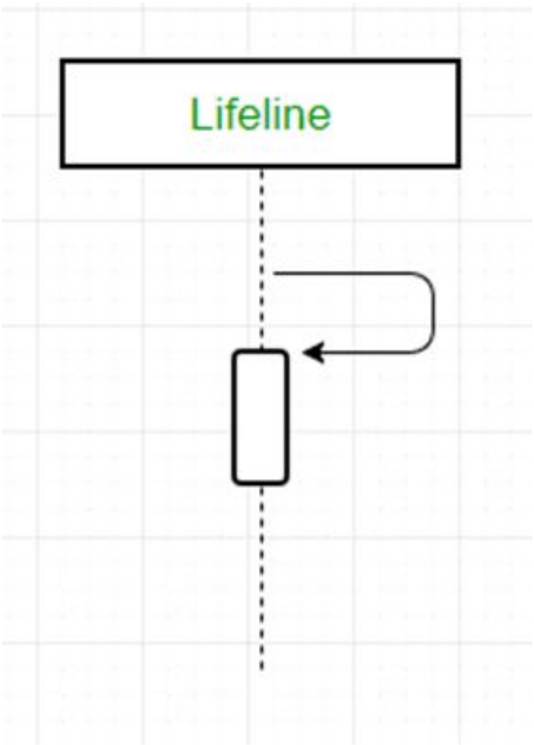
An Actor is an external user or system that starts the interaction with our system.

Actor = User (customer)

*The person who opens the app and clicks "**Place Order**".*

Sequence Diagram - Lifeline

A Lifeline represents an object or component that exists over time during the interaction.



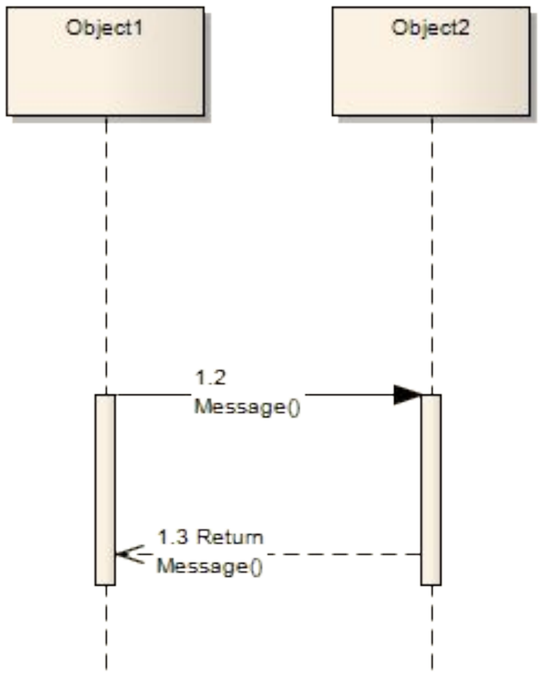
"Each lifeline is one vertical line in sequence diagram.
It answers: who is alive in this story?"

Sequence Diagram - Message

A Message is a method call or request sent from one lifeline to another.

- placeOrder()
- isOpen()
- save()

"Every arrow in sequence diagram is basically one method call between objects."



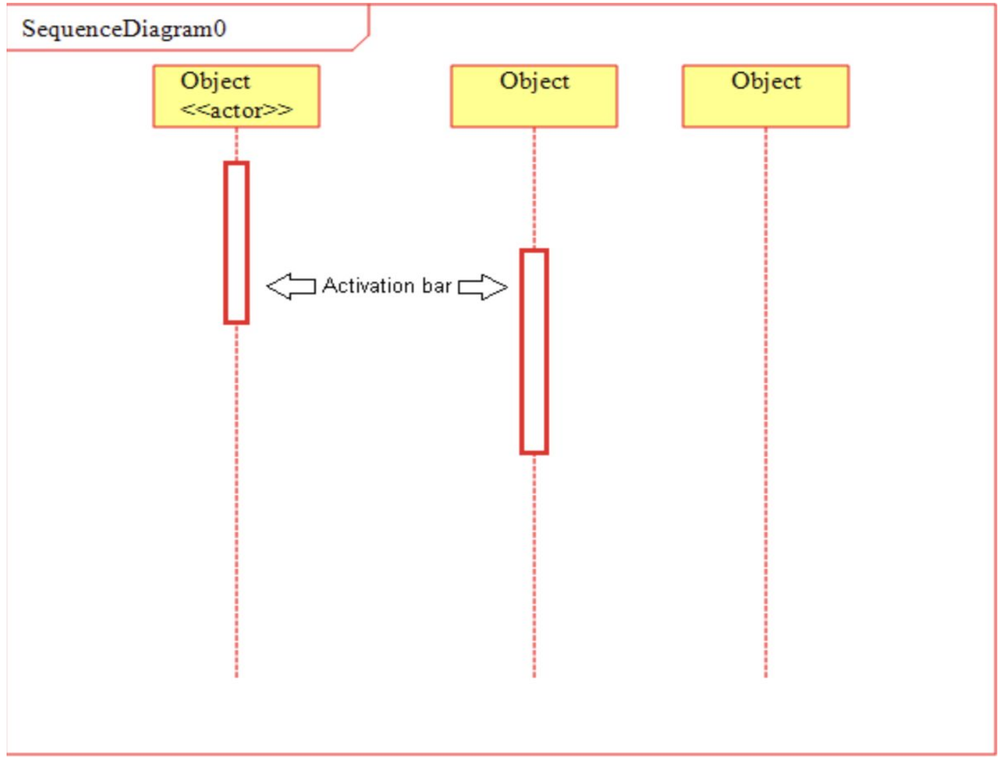
Sequence Diagram - Activity

An Activation bar shows the time period when an object is busy executing logic.

Example

- OrderService is busy:
- validating
- checking restaurant
- saving order

"Activation bar means:
this object is currently doing work."



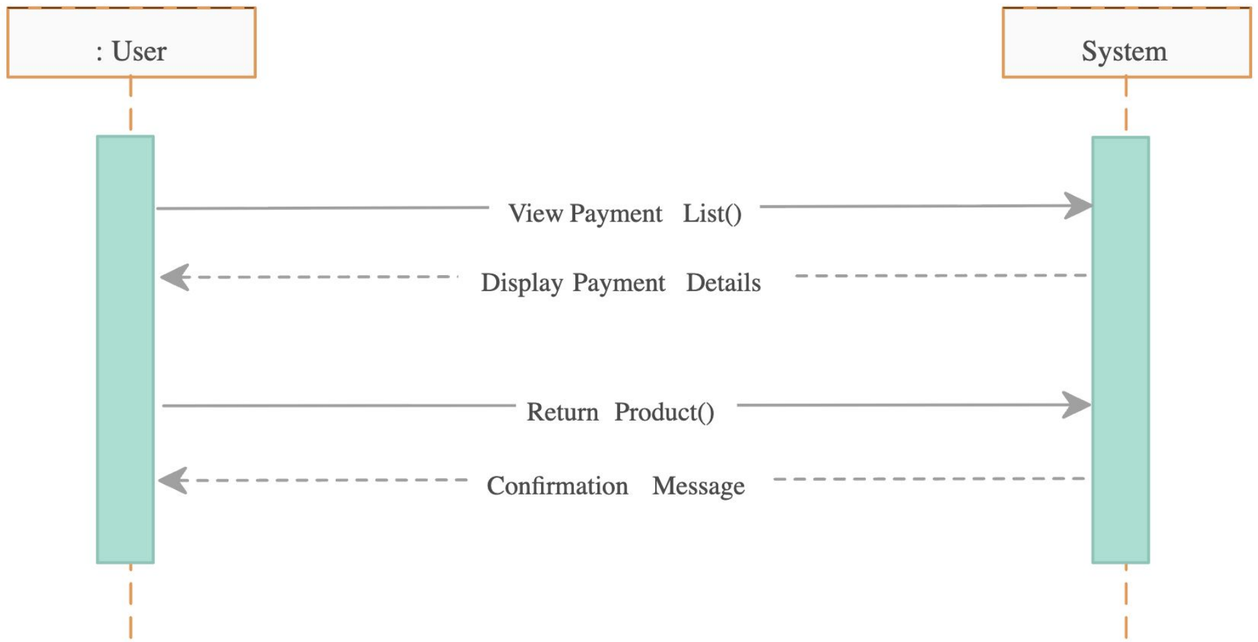
Sequence Diagram - Return

A Return message is the response sent back after a method finishes execution.
It is shown using a dashed arrow.

Example

RestaurantService returns: open/closed
DB returns: orderId

"Solid arrow = request
Dashed arrow = result"

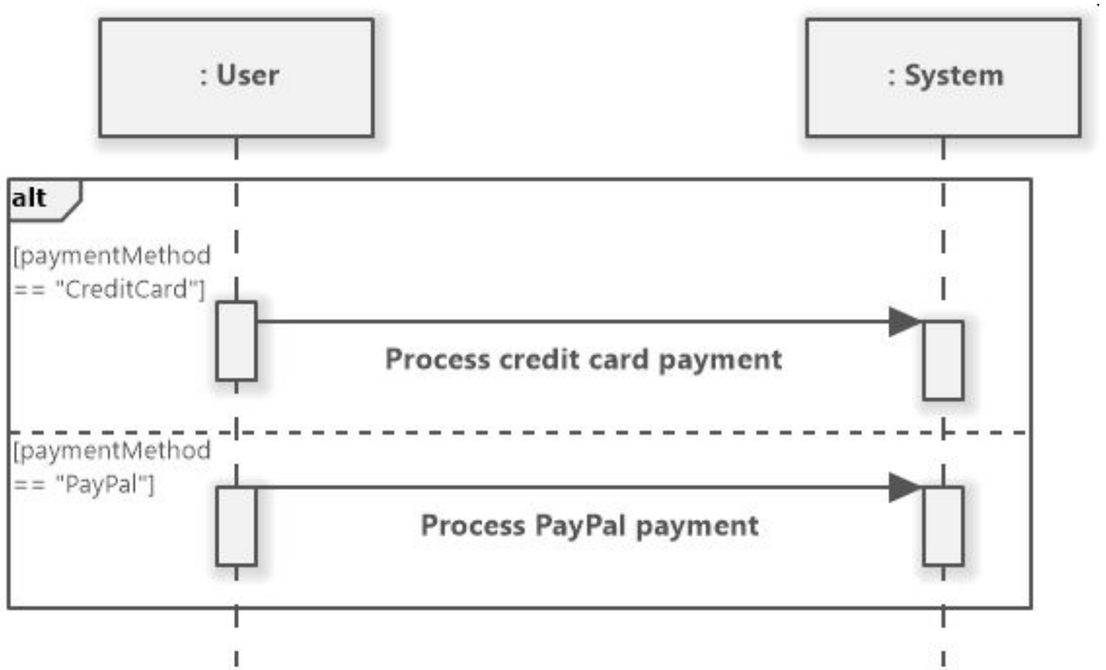


Sequence Diagram - Fragment: alt

Fragment: alt (if / else)

alt is used to show conditional paths (only one branch executes).

"alt is literally an if-else block in diagram form."



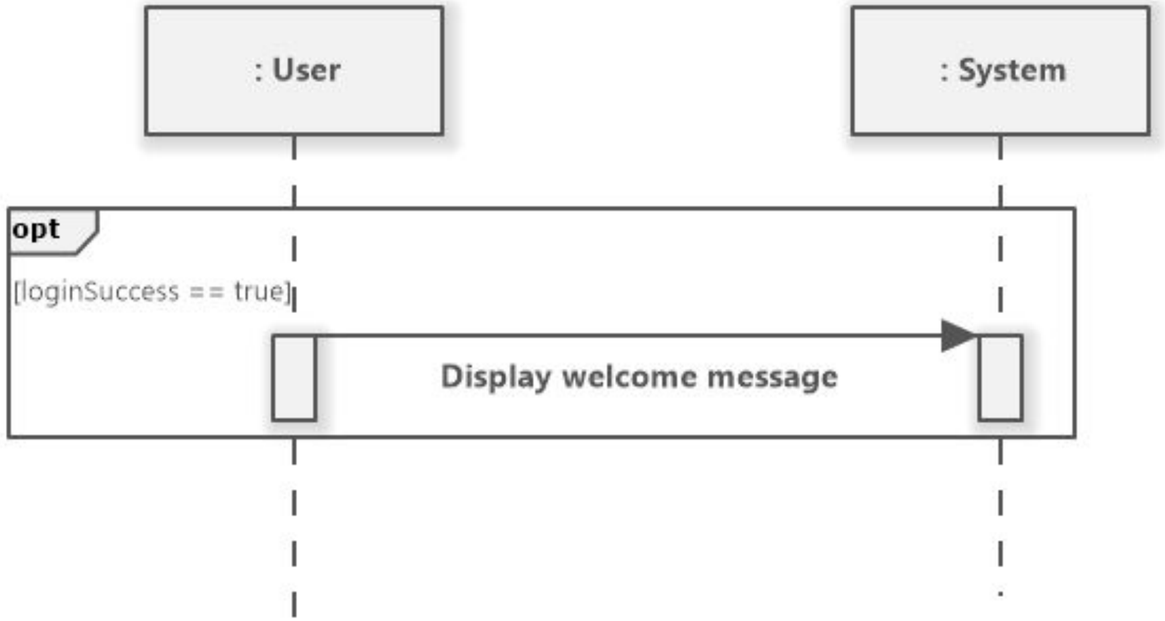
Sequence Diagram - Fragment: opt

Fragment: opt (optional)

opt is used for a step that may or may not happen.

Example

- Apply coupon (only if user entered code)
- Send confirmation email (optional)



""opt means:
this step is not guaranteed.""

Sequence Diagram Exercise

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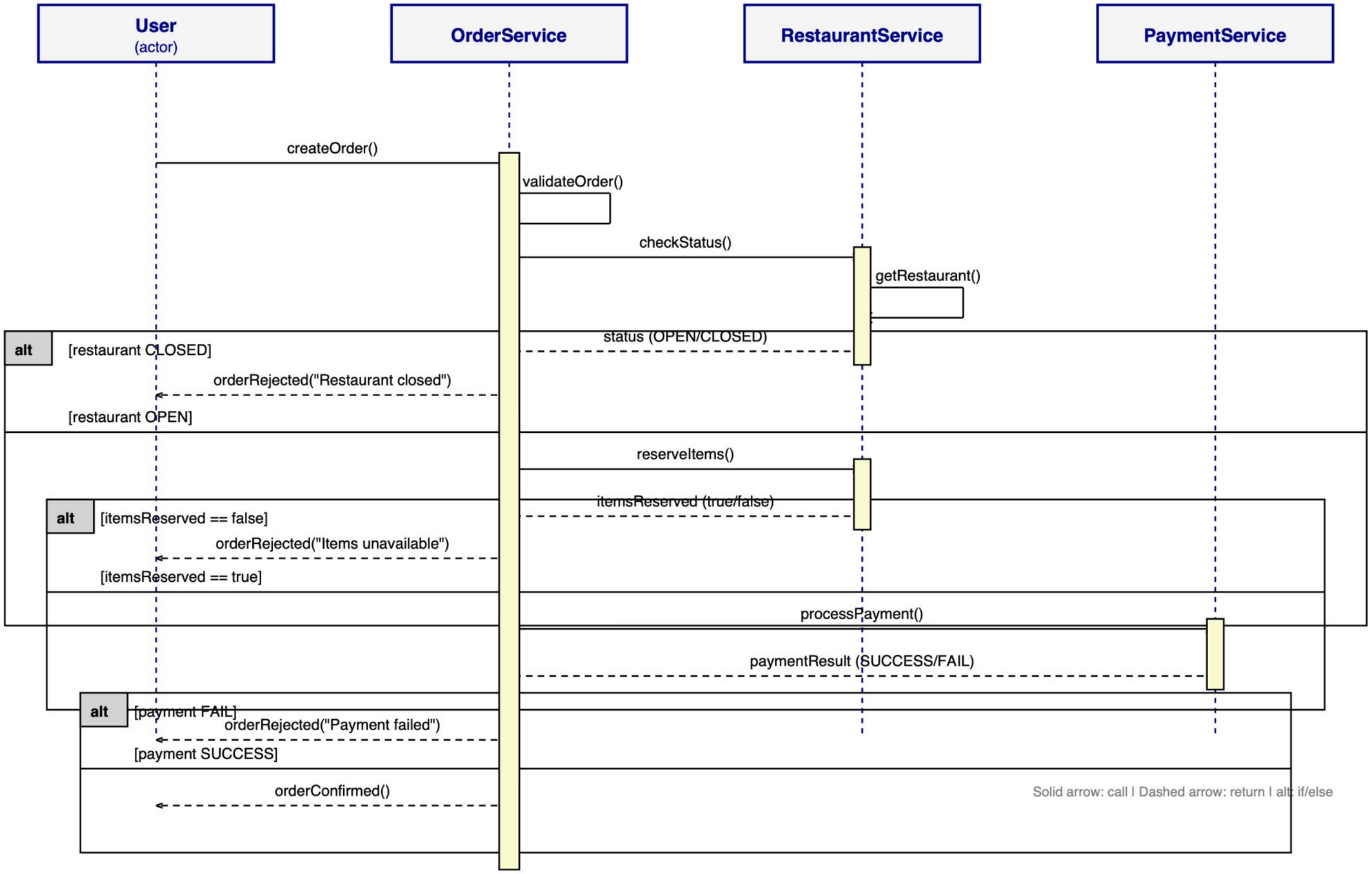
Create a sequence Diagram for a Food delivery app.

zomato SWIGGY


User
(actor)
OrderService
RestaurantService
PaymentService
createOrder()
validateOrder()
checkStatus()
getRestaurant()
status (OPEN/CLOSED)
alt
[restaurant CLOSED]
orderRejected("Restaurant closed")
[restaurant OPEN]
reserveItems()
itemsReserved (true/false)
alt
[itemsReserved == false]
orderRejected("Items unavailable")
[itemsReserved == true]
processPayment()
paymentResult (SUCCESS/FAIL)
alt
[payment FAIL]
orderRejected("Payment failed")
[payment SUCCESS]
orderConfirmed()
Solid arrow: call | Dashed arrow: return | alt: if/else

MCQ Time ..

Summary & Takeaways

- A class diagram tells what exists (structure), but it cannot tell what happens first/next (runtime order).
- A sequence diagram is the “runtime story” that removes ambiguity by showing who calls whom, in what order.
- Every sequence diagram is built from the same core blocks: Actor, Lifelines, Messages, Activations, Returns.
- alt is how we model real-world branching in flows (e.g., restaurant closed / payment failed), not extra text.
- If you can narrate a feature like Place Order step-by-step, you can convert that story directly into a correct UML sequence diagram.

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Thank You ..